

FUNDAMENTALS OF ELECTRICAL ENGINEERING

(BEE 201)

UNIT -1 : DC CIRCUITS

Lec-1

Today's Target

- *Basic definitions* ✓
- *Electric Circuit Elements* ✓
- *Concept of Energy Sources* ✓
- *University PYQs* ✓

Course Details(Paid) : All Subjects

✓ 1	Recorded Video Lectures (100 % Syllabus Coverage)
2	Pdf Notes
3	Lecture wise DPP
✓ 4	Unit wise set of PYQs

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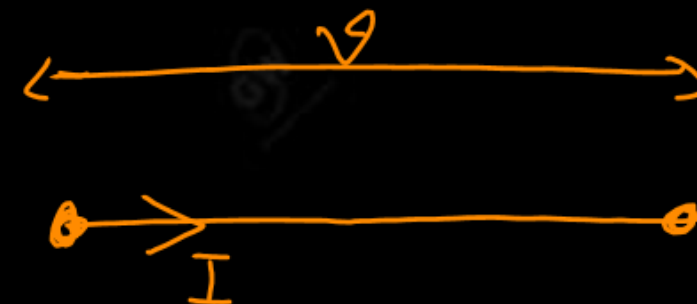
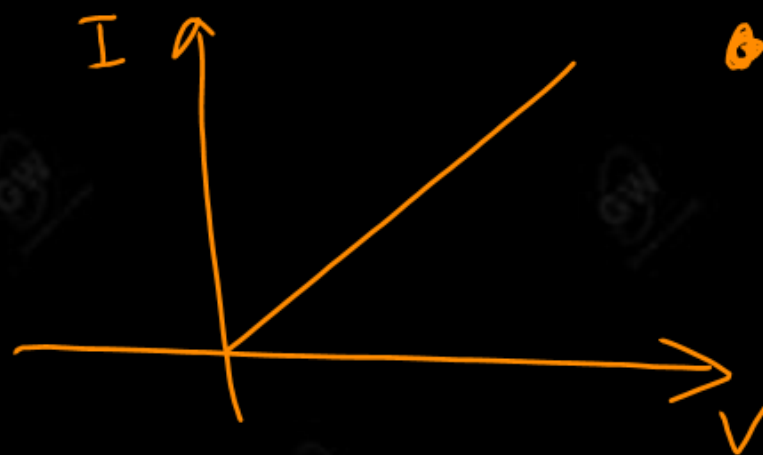
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Ohm's Law

Ohm's Law states that if all the physical conditions remains same then the current through a conductor is proportional to the voltage across the conductor.

$$V \propto I$$

$$V = IR$$



Elements of Electric Circuit

Resistor:- It is the opposition to the flow of current. It is a static element. The unit of resistance is ohm. Ω

$$R = \frac{V}{I}$$

$$R = \rho \frac{l}{a}$$

ρ → resistivity
 l → length of conductor
 a → c/s area of the conductor

Inductor:- An inductor is a passive electronic component that temporarily stores energy in a magnetic field when electric current flows through the inductor's coil. This is a dynamic element. The unit of inductor is Henry.

Voltage across inductor $V_L = L \frac{di}{dt}$



Capacitor:- A capacitor is defined as passive electric element that can store electric energy in the form of accumulated charges. This is also a dynamic element. The unit of capacitor is Farad.

Current across capacitor $i = C \frac{dv}{dt}$

$$Q = CV$$



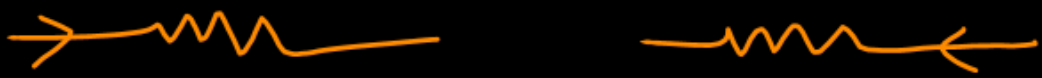
Basics of DC Circuits

(Imp for university Exams)

✓ **Active Elements:-** Elements which can provide energy to the circuits are known as active elements. Ex. Battery

✓ **Passive Elements:-** Elements which can dissipate or store energy are known as passive elements. Ex. Resistor, Capacitor and Inductor

Bilateral Element:- Elements whose characteristic does not change with the direction of current are known as bilateral elements. Ex. Resistor, Capacitor, Inductor etc



Unilateral Element:- Elements whose characteristic changes with the direction of current are known as unilateral elements. Ex. Diode, Transistor etc

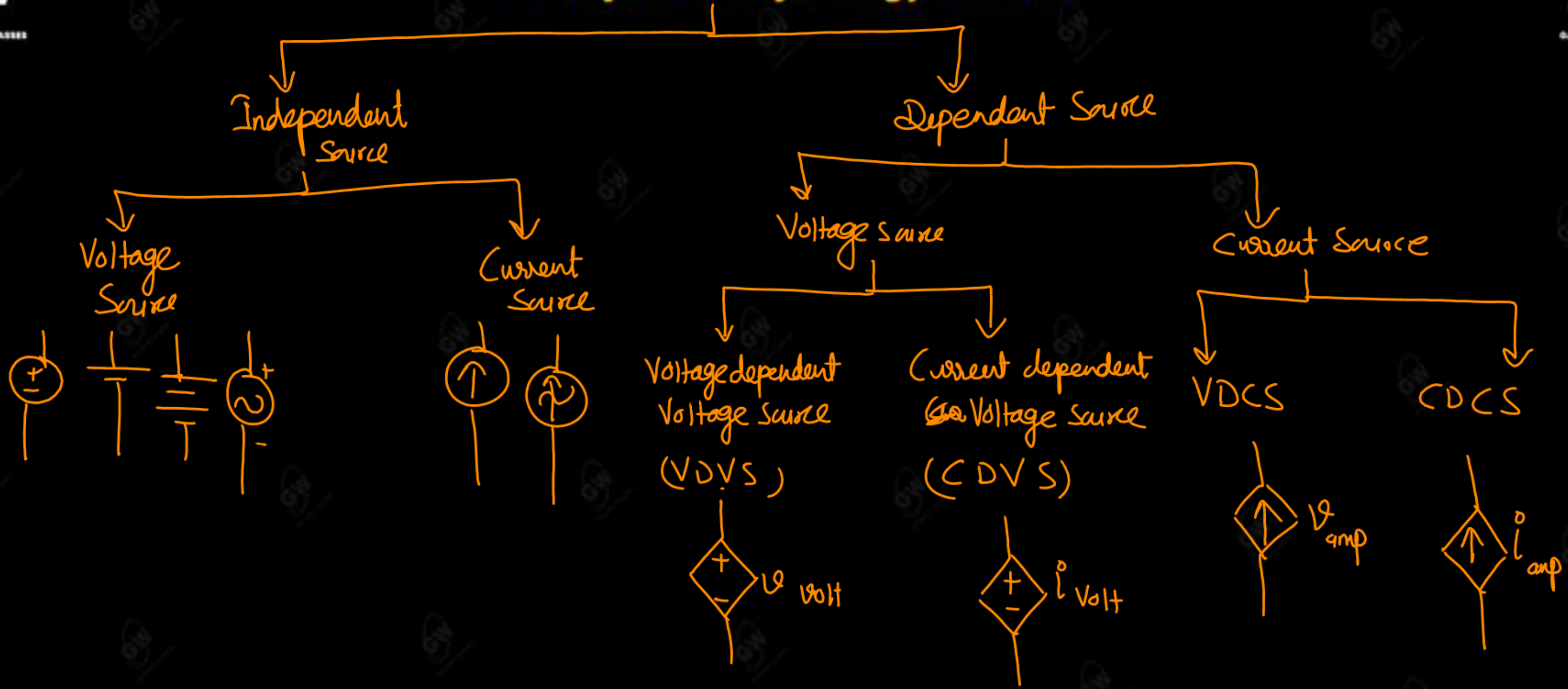


Linear and Non Linear Elements:- Linear elements are the elements which shows linear characteristic between voltage and current Also follows the principle of superposition and homogeneity Ex- R, L, C

Non-linear → diode

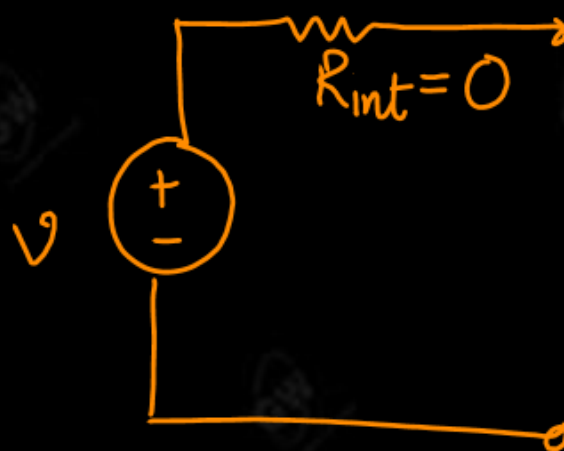
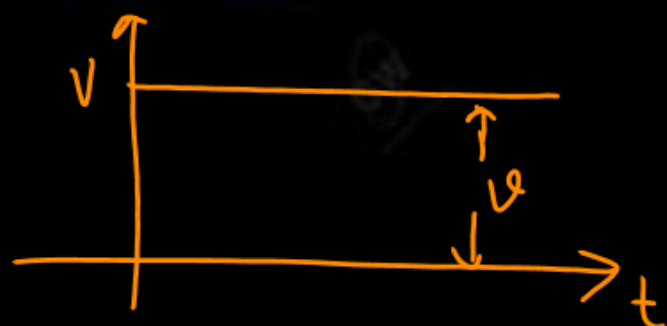


Classification of Energy Sources

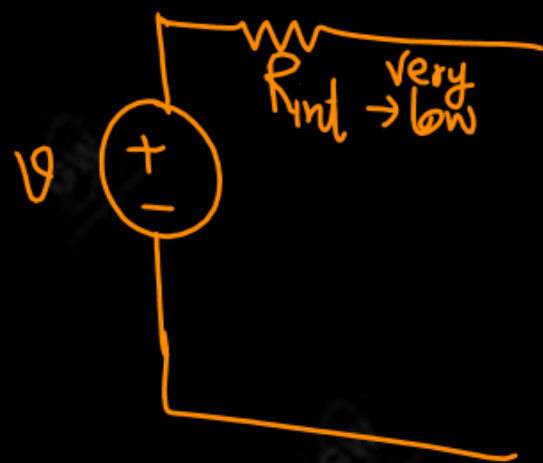


Voltage Source

Ideal Voltage source:- Voltage source that provide constant voltage to the load is known as ideal voltage source.

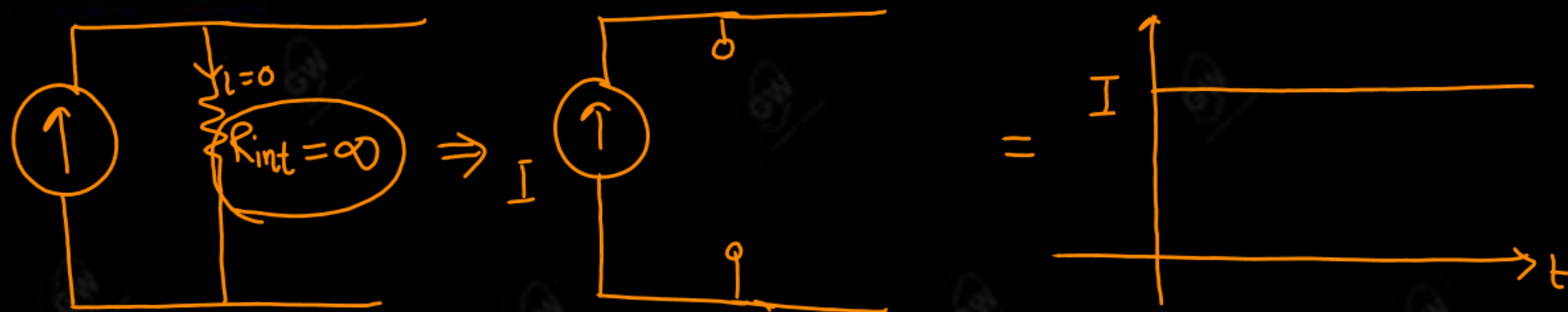


Practical Voltage Source:- Practical voltage sources are real and used in daily life. A practical voltage source has internal resistance that causes a drop in terminal voltage due to the current flow.

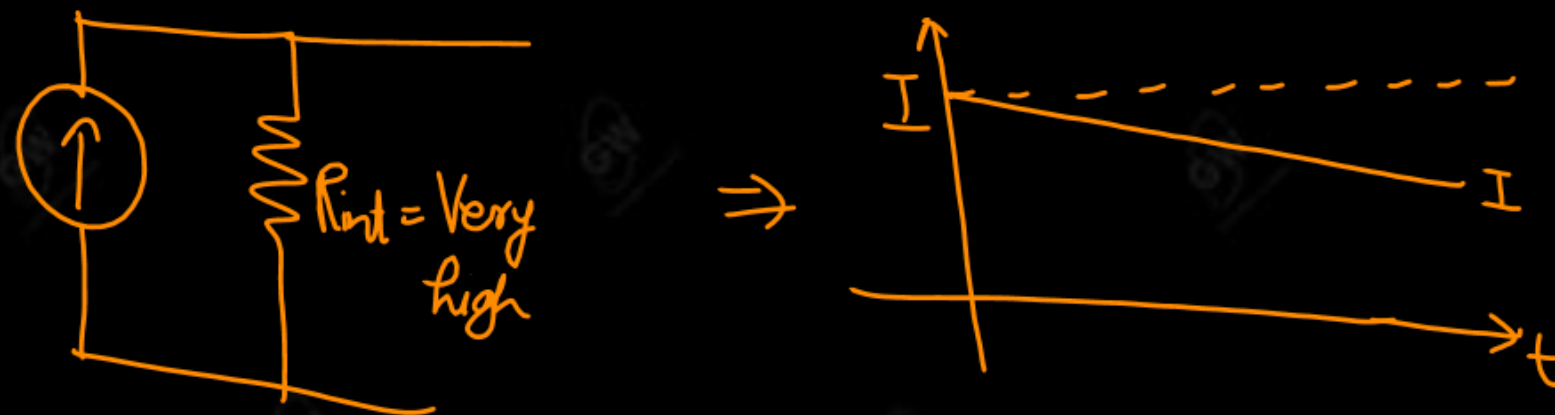


Current Source

Ideal Current source: Current source that provide constant voltage to the load is known as ideal Current source.



Practical Current Source: Practical Current sources are real and used in daily life. A practical current source has internal resistance that causes a drop in load current.



Basics Terminology of DC Circuits

(BCDE)
(ABEF)
(ABCDEF)

Loop And Mesh:- A loop is any closed path thorough a circuit. While a mesh is a loop that does not contain any other loop in it.

(ABEF)
(BCDE)

every loop $\rightarrow$ mesh ✗
every mesh $\rightarrow$ loop ✓

A, B, C, E

Node and Junction:- A node is a point where two or more elements can meet. While a junction is point where three or more points can meet.

(B, E)

every Junction $\rightarrow$ node ✓
every node $\rightarrow$ Junction ✗

Electric Circuit and Electric Network:- A circuit is a closed conducting path through which an electric current either flow or is intended flow While A combination of various electric elements, connected in any manner. Whatsoever, is called an electric network

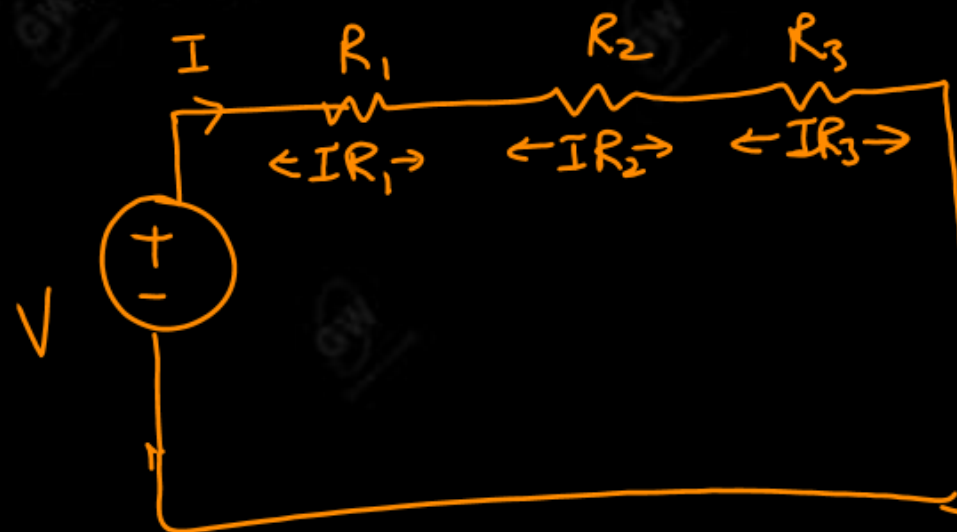


Kirchhoff's Voltage Law (KVL)

According to KVL algebraic Sum of all voltages in a loop is always zero.

$$\sum V = 0$$

$$V - I(R_1 + R_2 + R_3) = 0$$



Kirchhoff's Currents Law (KCL)

According to KCL algebraic Sum of all currents at a node is always zero.

Or

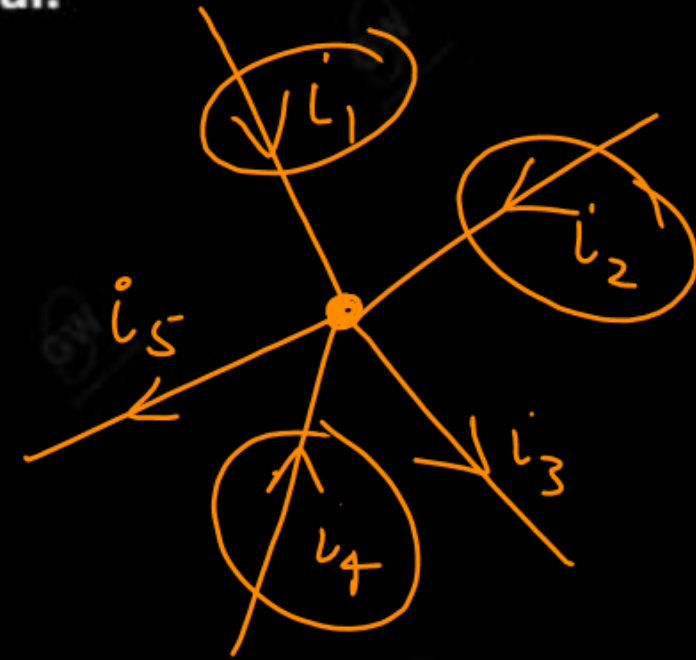
Sum of incoming current and sum of outgoing current at a node is always equal.

$$\sum I = 0$$

$$i_1 + i_2 + i_4 - i_3 - i_5 = 0$$

$$(i_1 + i_2 + i_4) = (i_3 + i_5)$$

$$\begin{aligned} I_{in} &\rightarrow +ve \\ I_{out} &\rightarrow -ve \end{aligned}$$



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Thank You

FUNDAMENTALS OF ELECTRICAL ENGINEERING

(BEE 201)

UNIT -1 : DC CIRCUITS

Lec-2

Today's Target

- *Method of solving circuit by Mesh analysis*
- *University PYQs on Mesh Analysis*
- *DPP*

Mesh Analysis(KVL)

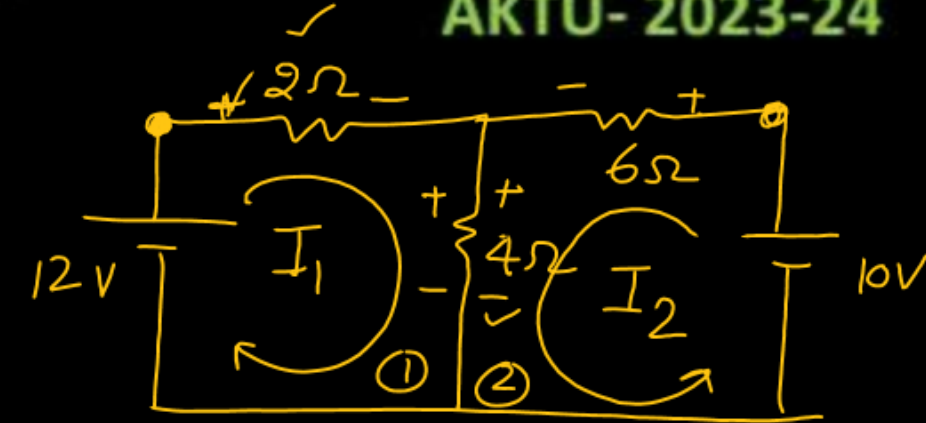
$$\rightarrow \text{---} \overset{+}{\text{m}} \text{---} \quad V = IR$$

Explain the procedure of mesh analysis with the help of an example.

Steps →

- ① → find mesh in the given circuit
- ② Assume loop current in the mesh
- ③ Now apply KVL in each loop, we will get one equation from each loop
- ④ Now Solve these equations to find loop currents

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$$+ \underbrace{2I_1} + \underbrace{4(I_1 + I_2)} - 12 = 0$$

$$6I_1 + 4I_2 = 12 \quad \text{--- ①}$$

$$3I_1 + 2I_2 = 6 \quad \text{--- ②} \times 5$$

$$6I_2 + 4(I_1 + I_2) - 10 = 0$$

$$4I_1 + 10I_2 = 10 \quad \text{--- ②}$$

$$-15I_1 + 10I_2 = 30$$

$$+ 11I_1 = +20$$

$$\boxed{I_1 = 20/11}$$

Methods for solving electric Network Using KVL (Mesh Analysis)

Ex.- Using Mesh Analysis Find Current in all branches.

1. Loop Current Method

Apply KVL in loop ①

$$+1I_1 + 6(I_1 + I_2) + 2I_1 = 18$$

$$9I_1 + 6I_2 = 18 \quad \text{--- ①}$$

Apply KVL in loop ②

$$+3I_2 + 6(I_2 + I_1) + 4I_2 = 12V$$

$$6I_1 + 13I_2 = 12 \quad \text{--- ②}$$

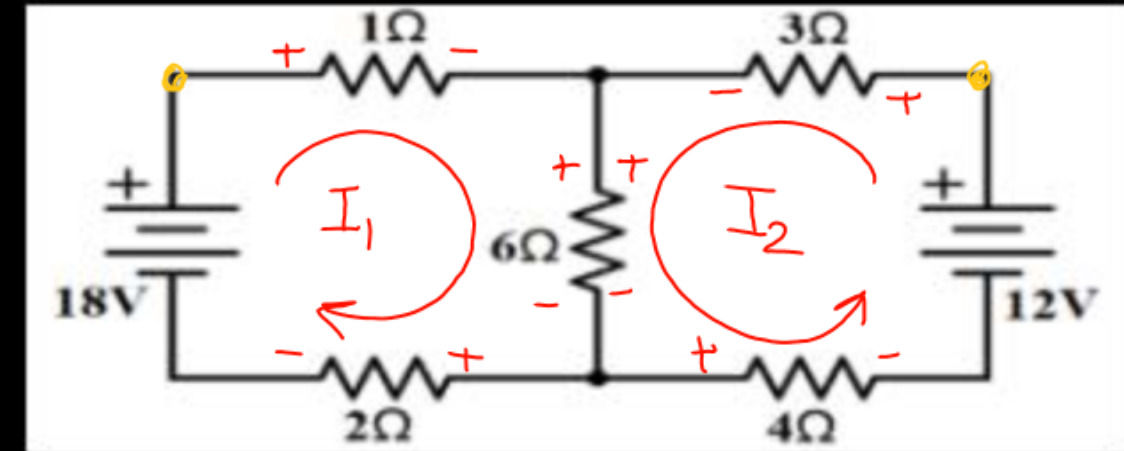
Now solve Eq ① ②

$$I_1 = 2 \text{ amp}$$

$$I_2 = 0 \text{ A}$$

Ans

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Kirchhoff's Voltage Law (KVL)

Ex.- Using Mesh Analysis Find Current in all branches.

1. Branch Current Method

Apply KVL in loop ①

$$+I_1 + 6(I_1 - I_2) + 2I_1 = 18$$

$$9I_1 - 6I_2 = 18 \quad \text{--- (1)}$$

Apply KVL in loop ②

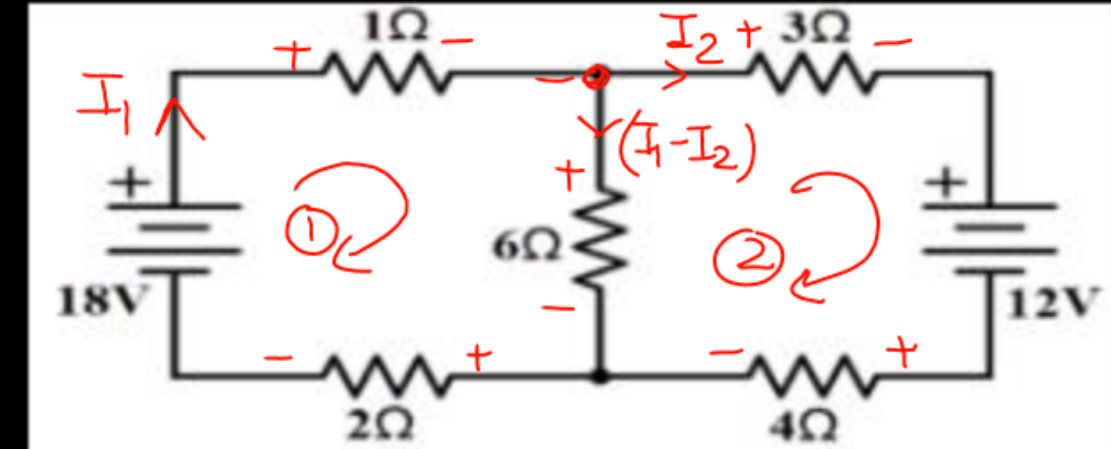
$$+3I_2 + 4I_2 - 6(I_1 - I_2) = 0$$

$$-6I_1 + 13I_2 = 0 \quad \text{--- (2)}$$

Now Solve Eqn ① & ②

$$\left. \begin{array}{l} I_1 = 2 \text{ amp} \\ I_2 = 0 \end{array} \right\}$$

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Kirchhoff's Voltage Law (KVL)

Ex.- Using Mesh Analysis Find Current in all branches.

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Apply KVL in Loop ②

$$5I + 2(I + I_1) + 2(I - 2) = 0$$

$$9I + 2I_1 = 4 \quad \text{--- (1)}$$

Apply KVL in Loop ③

$$+4I_1 + 2(I_1 + I) = 100$$

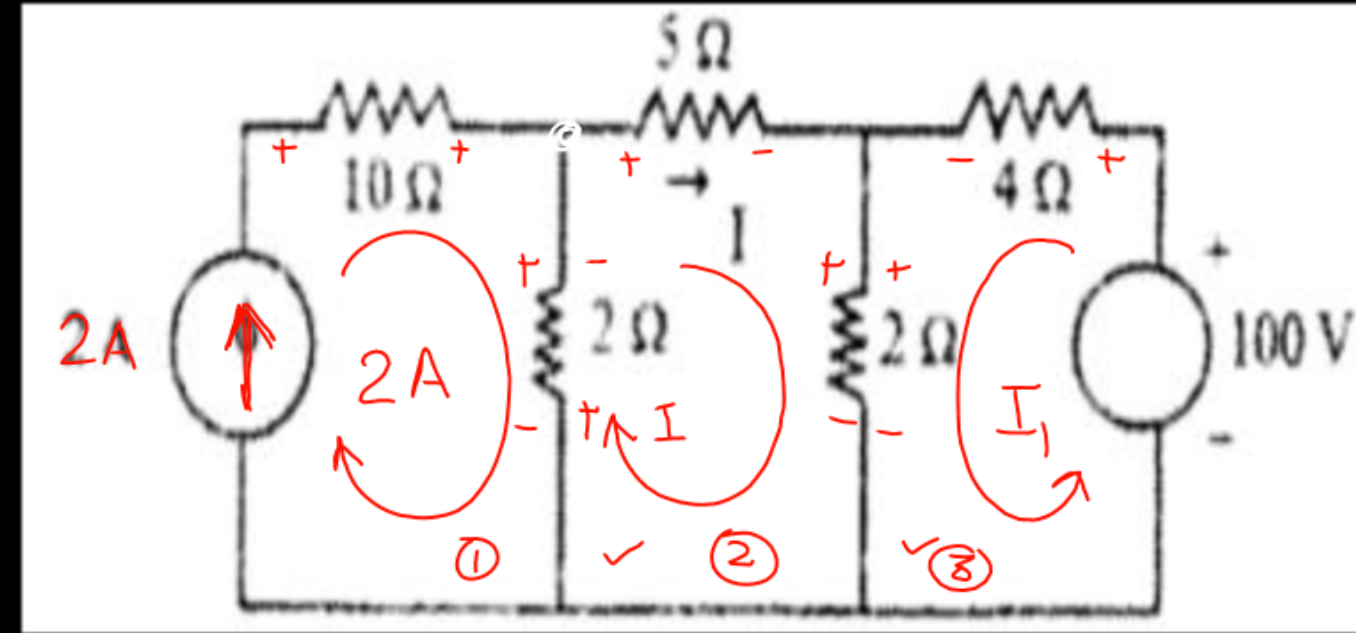
$$2I + 6I_1 = 100 \quad \text{--- (2)}$$

Solve Equ (1) & (2)

$$I = -3.52 \text{ A}$$

$$I_1 = 17.84 \text{ A}$$

Ans



Mesh Analysis

Ques. Find Current in 1 ohm Resistance using Mesh Analysis.

Apply KVL in loop ①

$$-2(I-1) + 1(I+3) = 4$$

$$3I - 2 + 3 = 4$$

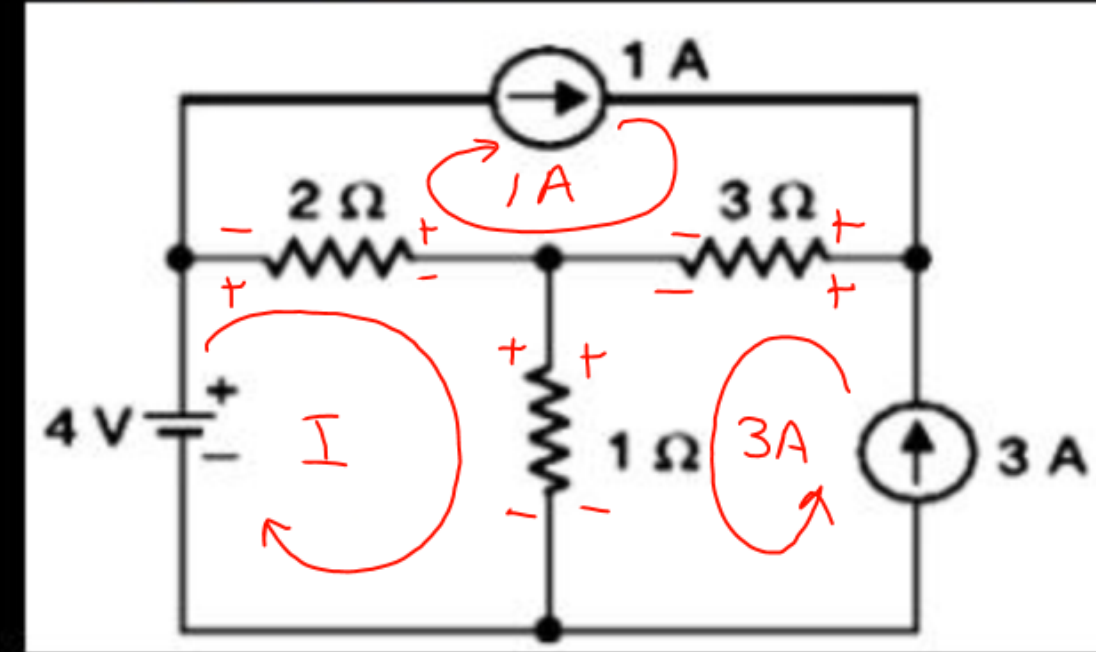
$$3I = 3$$

$$I = \frac{3}{3}$$

$$I = 1 \text{ amp}$$

Ans

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Ex. Using Mesh Analysis Find Current in 6 ohm Resistance.

Apply KVL in Loop ①

$$+3I_1 + 6(I_1 - I_2) = 6$$

$$9I_1 - 6I_2 = 6 \quad \text{--- ①}$$

Apply KVL in Supermesh

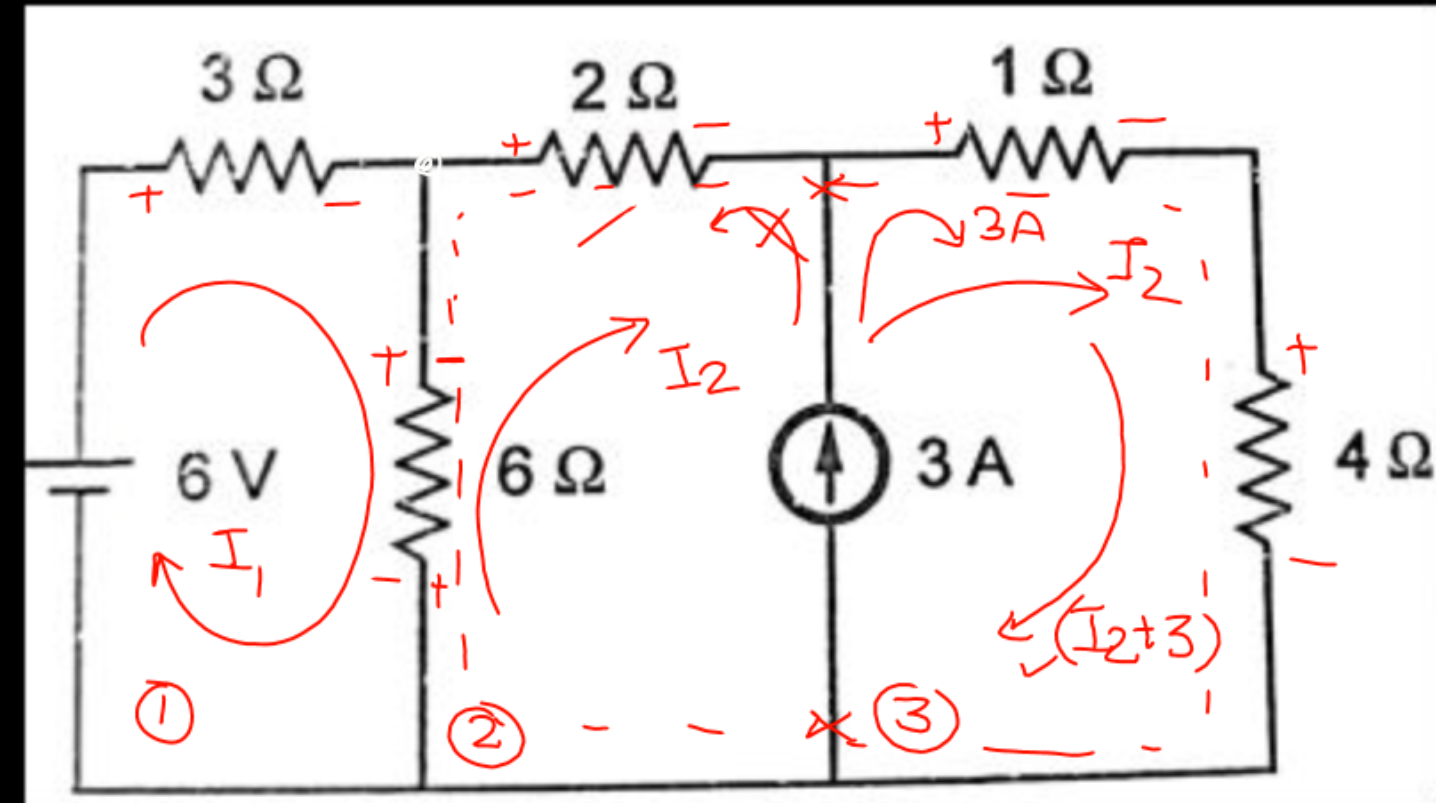
$$+2I_2 + 1(I_2 + 3) + 4(I_2 + 3) + 6(I_2 - I_1) = 0$$

$$2I_2 + 5(I_2 + 3) + 6I_2 - 6I_1 = 0$$

$$13I_2 + 15 - 6I_1 = 0$$

$$-6I_1 + 13I_2 = -15 \quad \text{--- ②}$$

Now solve Eqn ① & ②



$$I_1 = -0.14 \text{ A}$$

$$I_2 = -1.2 \text{ A}$$

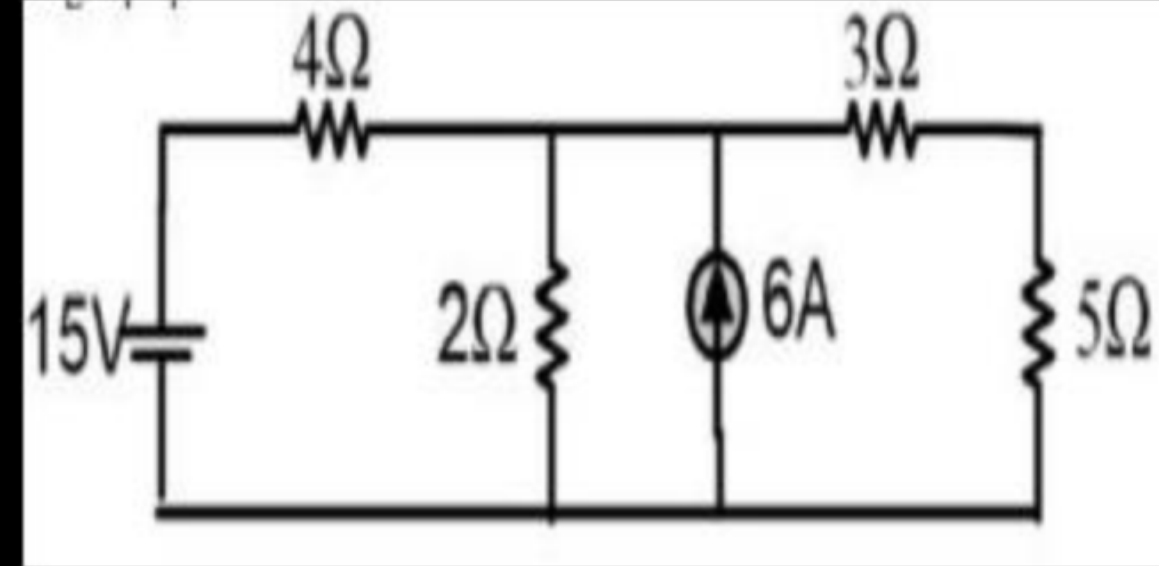
$$I_{6\Omega} = (I_2 - 6) = -1.2 - 6 = -7.2 \text{ A} \uparrow$$

$$= 7.2 \text{ A} \downarrow$$

Q.1 Using Mesh Analysis Find Current in 5 Ohm.

Ans:- $I = 1.4A$

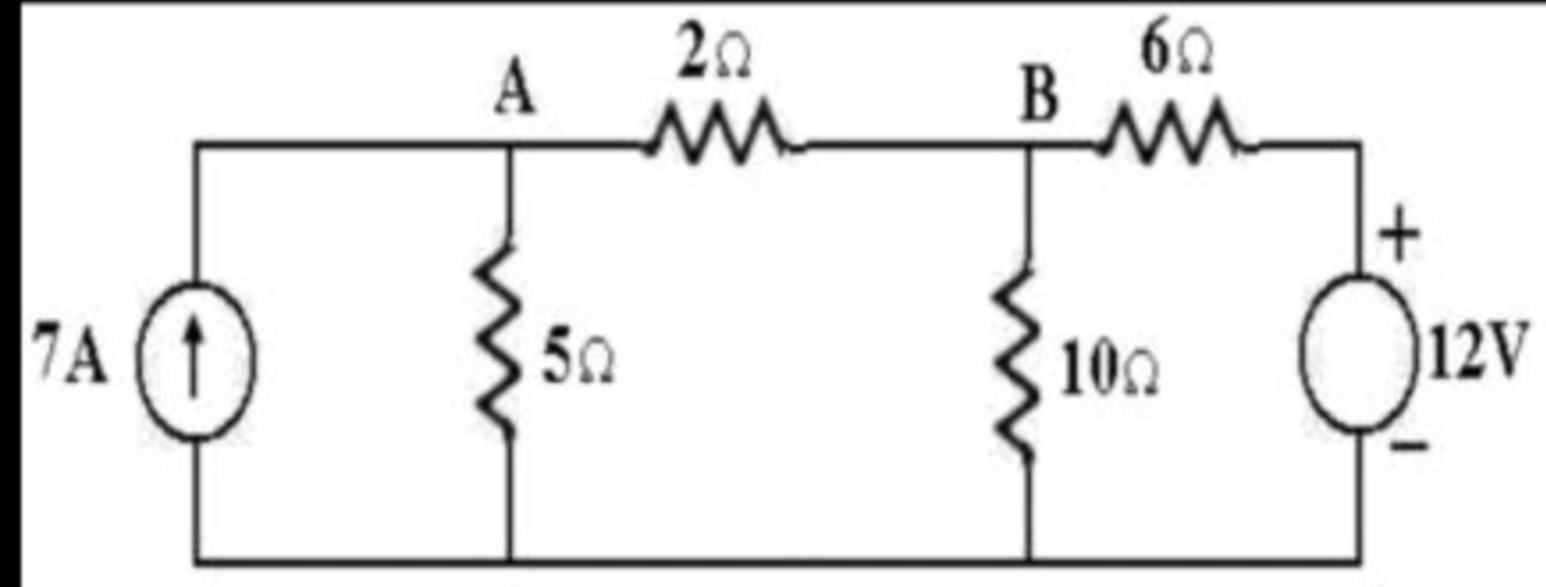
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Q.2 Using Mesh Analysis Find Current in 2 Ohm.

Ans:- $I = 2.55 A$

(2023-24)



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Thank You

B.Tech First Year

ELECTRICAL ENGINEERING

UNIT DC CIRCUITS

Lec-3

Today's Target

- ✓ Examples on Nodal Analysis/KVL ✓
- ✓ Previous Year Questions ✓
- ✓ DPPs ✓

Process to Apply Nodal Analysis

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Steps →

① Identify Junctions (Node) in the given Circuit.

② Assume reference Node (Ground)

③ Label remaining Nodes

④ Assume branch Current

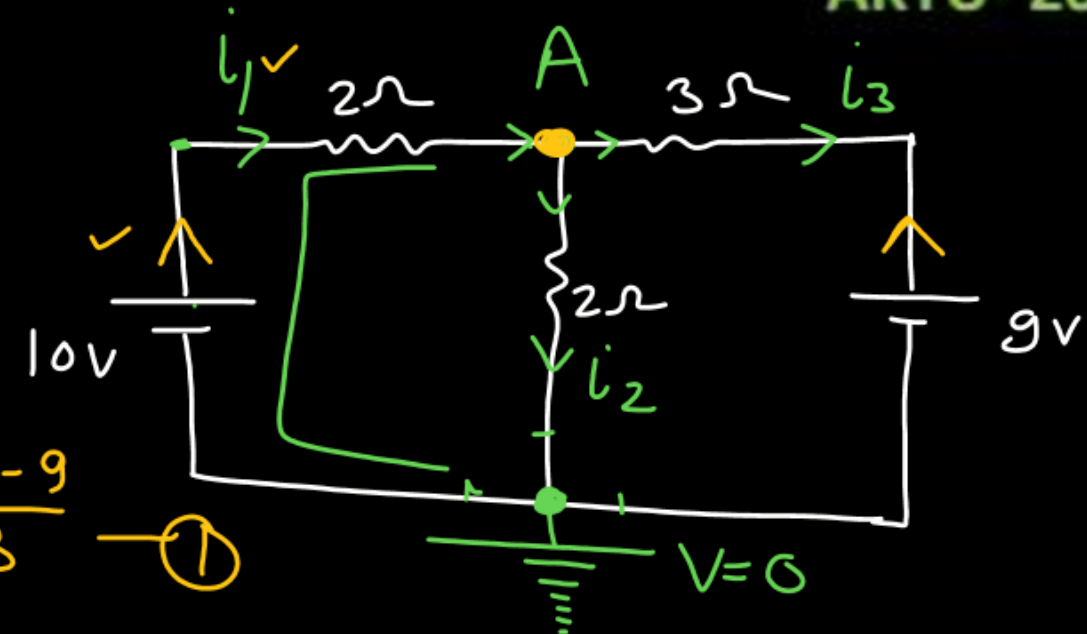
⑤ Write branch Current in terms of Node Voltage

⑥ Apply KCL at each node and make Equations in terms of Node Voltage

⑦ Solve these equations to find Node Voltages

$$i_1 = i_2 + i_3$$

$$\frac{10 - V_A}{2} = \frac{V_A}{2} + \frac{V_A - 9}{3}$$



Junction → where 3 or more elements meet

(Current distribution)

$$i_1 = \frac{(0 - V_A + 10)}{2}, \quad i_2 = \frac{V_A - 0}{2}$$

$$i_3 = \frac{V_A - 9}{3}$$

Ex-1 Using Nodal Analysis Find Current in all branches.

Solⁿ

$$i_1 = \frac{V_A - 0}{2} = \frac{V_A}{2} \quad ; \quad i_4 = \frac{V_B}{5}$$

$$i_2 = \frac{V_A - V_B}{3} \quad ; \quad i_5 = \frac{(V_B - V_C)}{1}$$

$$i_3 = \frac{V_A - V_C}{5} \quad ; \quad i_6 = \frac{V_C}{4}$$

Apply KCL at Node 'A', we get

$$i_1 + i_2 + i_3 = 10$$

$$\frac{V_A}{2} + \frac{V_A - V_B}{3} + \frac{V_A - V_C}{5} = 10$$

$$V_A \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{5} \right) - \frac{V_B}{3} - \frac{V_C}{5} = 10$$

$$31 \frac{V_A}{30} - \frac{V_B}{3} - \frac{V_C}{5} = 10 \quad \text{--- (1)}$$

Apply KCL at Node B

$$i_2 = i_4 + i_5$$

$$i_2 - i_4 - i_5 = 0$$

$$\frac{V_A - V_B}{3} - \frac{V_B}{5} - (V_B - V_C) = 0$$

$$\frac{V_A - V_B}{3} - \frac{V_B}{5} - (V_B - V_C) = 0$$

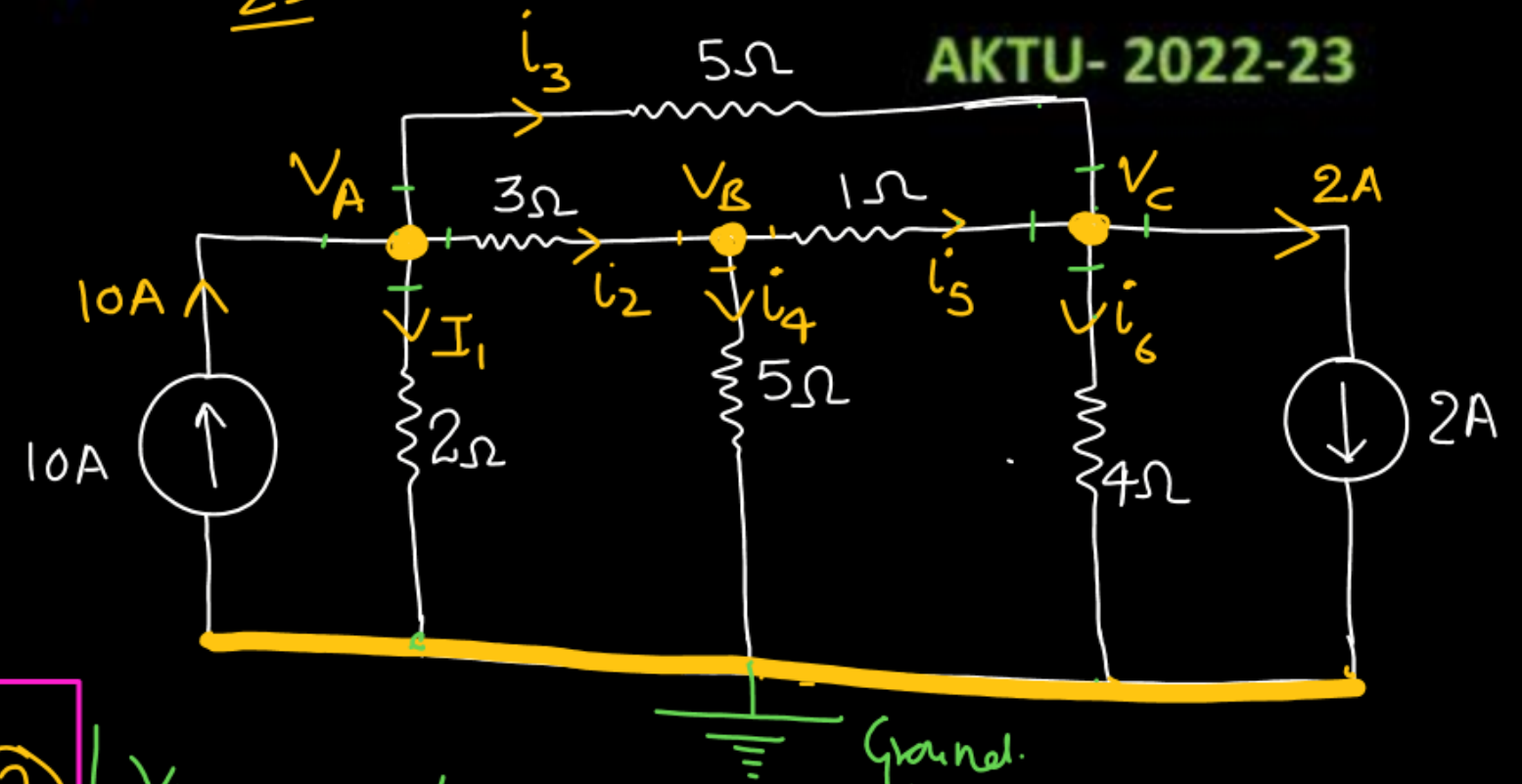
$$\frac{V_A}{3} - \frac{23}{15} V_B + V_C = 0 \quad \text{--- (2)}$$

Apply KCL at Node C

$$i_3 + i_5 = i_6 + 2$$

$$i_3 + i_5 - i_6 = 2$$

$$\frac{V_A - V_C}{5} + V_B - V_C - \frac{V_C}{4} = 2$$



$$\frac{V_A}{5} + V_B - V_C \left(\frac{1}{5} + 1 + \frac{1}{4} \right) = 2$$

$$\frac{V_A}{5} + V_B - \frac{29}{20} V_C = 2 \quad \text{--- (3)}$$

Solve Equⁿ (1), (2) & (3)

$$V_A = 12.05 \text{ Volt}$$

$$V_B = 5.10 \text{ Volt}$$

$$V_C = 3.80 \text{ Volt}$$

$$i_1 = \frac{V_A}{2} = \frac{12.05}{2} = 6.0 \text{ A}$$

$$i_2 = \frac{12.05 - 5.10}{3} = 2.3 \text{ A}$$

$$i_3 = \frac{12.05 - 3.80}{5} = 1.65 \text{ A}$$

Ans

Nodal Analysis

Ex.2- Using Nodal Analysis Find Current in 20 ohm resistor all branches.

$$i_1 = \frac{20 - V_A}{5} = \frac{20 - 20}{5} = 0$$

$$i_2 = \frac{V_A}{10} = \frac{20}{10} = 2 \text{ amp}$$

$$i_3 = \frac{V_A - V_B}{10} = \frac{20 - 40}{10} = -2 \text{ amp}$$

$$i_4 = \frac{V_B}{20} = \frac{40}{20} = 2 \text{ amp} \quad \text{Ans}$$

Apply KCL at Node A,

$$i_1 = i_2 + i_3$$

$$i_1 - i_2 - i_3 = 0$$

$$\frac{20 - V_A}{5} - \frac{V_A}{10} - \frac{(V_A - V_B)}{10} = 0$$

$$-V_A \left(\frac{1}{5} + \frac{1}{10} + \frac{1}{10} \right) + \frac{V_B}{10} = -4$$

$$-V_A \left(\frac{4}{10} \right) + \frac{V_B}{10} = -4 \quad \text{--- (1)}$$

Apply KCL at Node B

$$i_3 + 4 = i_4$$

$$i_3 - i_4 = -4$$

$$\frac{V_A - V_B}{10} - \frac{V_B}{20} = -4$$

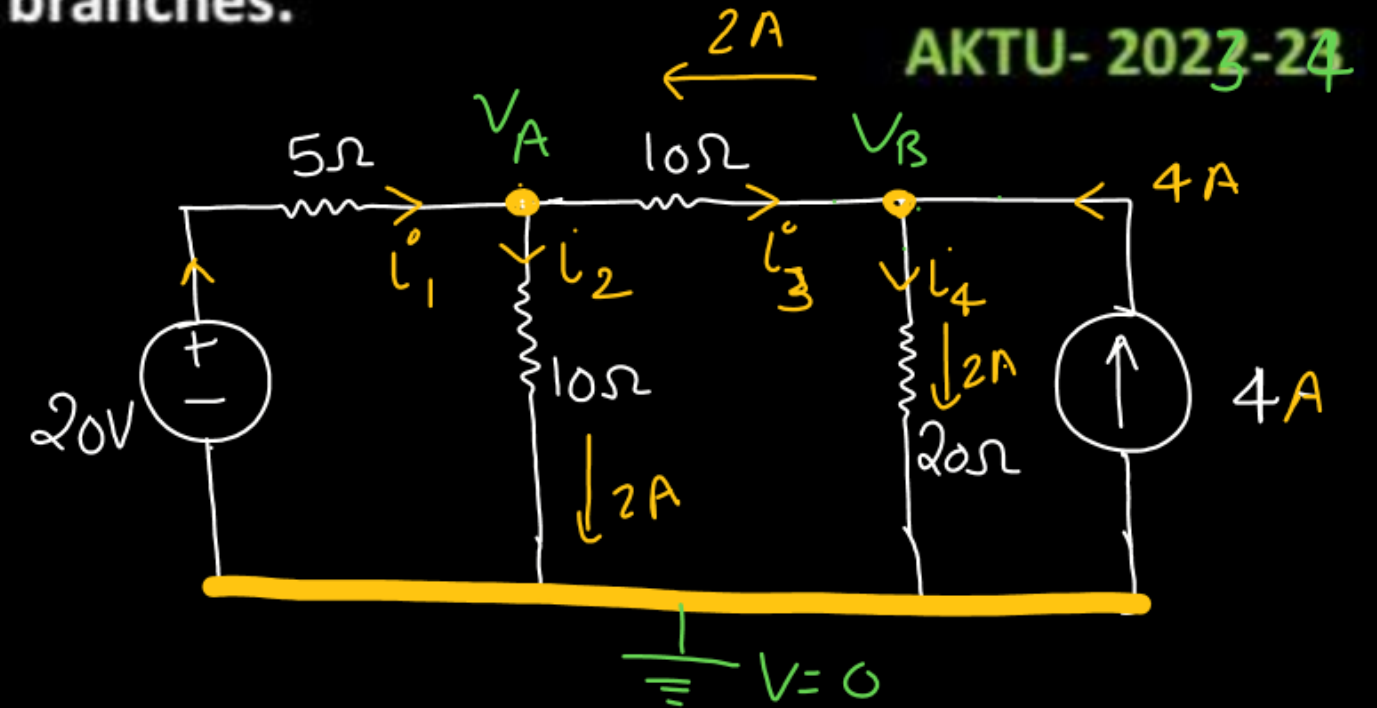
$$\frac{V_A}{10} - V_B \left(\frac{1}{10} + \frac{1}{20} \right) = -4$$

$$\frac{V_A}{10} - \frac{3}{20} V_B = -4 \quad \text{--- (2)}$$

Solve Equⁿ (1) & (2)

$$V_A = 20 \text{ V}$$

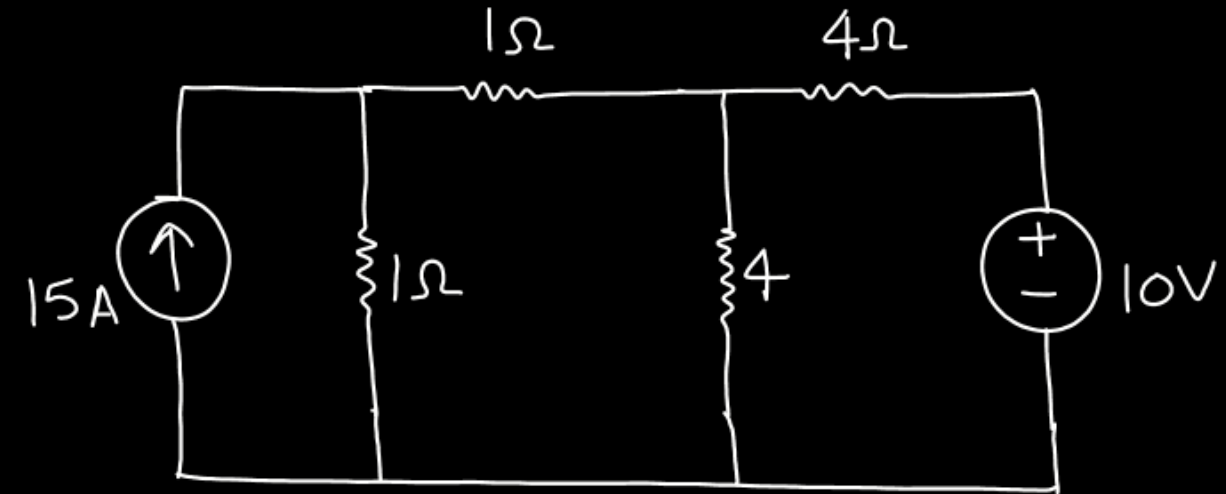
$$V_B = 40 \text{ V}$$



Ex.3- Using Nodal Analysis Find Current in all branches. **AKTU- 2021-22**

Similar as Ques ②

Do practice at Home



Nodal Analysis

Ex-4 Using Nodal Analysis Find Current in 8 ohm Resistance.

$$i_1 = \frac{0 - V_A + 2}{6} = \frac{2 - V_A}{6}$$

$$i_2 = \frac{V_A}{8}$$

Apply KCL at Node A,

$$i_1 + 2 = i_2$$

$$i_1 - i_2 = -2$$

$$\frac{2 - V_A}{6} - \frac{V_A}{8} = -2$$

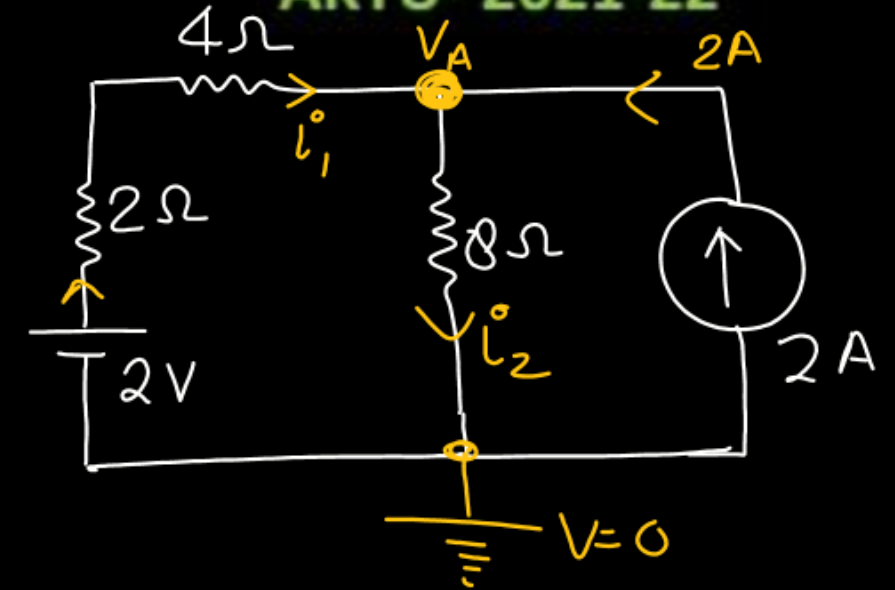
$$\frac{1}{3} - \frac{V_A}{6} - \frac{V_A}{8} = -2$$

$$+ V_A \left(\frac{1}{6} + \frac{1}{8} \right) = +2.33$$

$$V_A = 8 \text{ Volt}$$

$$i_2 = \frac{8}{8} = 1 \text{ amp}$$

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Concept of Super Node

Ex.5 Using Nodal Analysis Find Current in 2 ohm Resistance.

Super Node $\rightarrow$ When a voltage source is connected b/w two nodes without any resistance. In that case we consider both nodes as a single node known as Super node.

Apply KCL at Supernode

$$i_1 + i_2 + 2 = 3$$

$$i_1 + i_2 = 1$$

$$\frac{(V_A - 0)}{2} + \frac{(V_B - 0)}{1} = 1$$

$$\boxed{\frac{V_A}{2} + V_B = 1} \quad \text{--- ①}$$

from Supernode

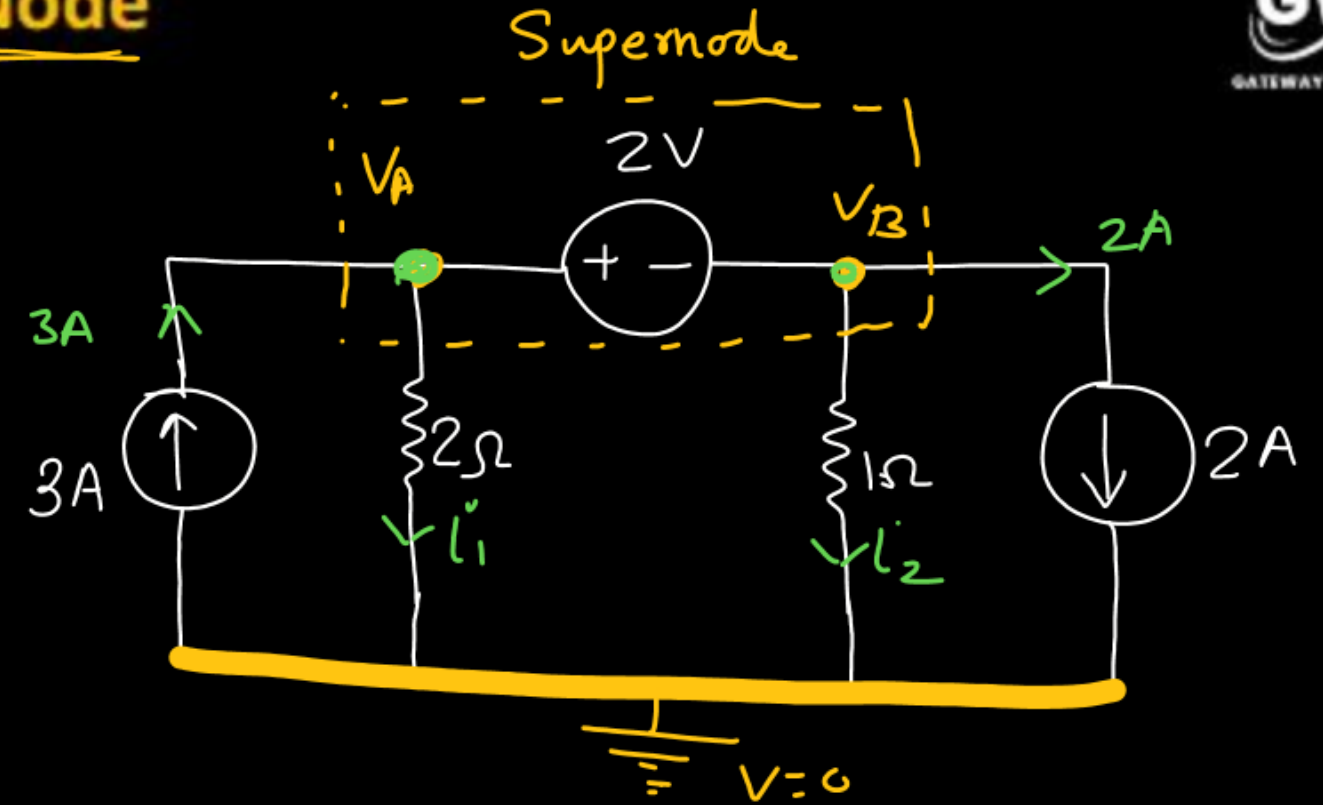
$$V_A - V_B = 2 \quad \text{--- ②}$$

Now solve Eqs ① & ②

$$V_A = 2 \text{ volt}$$

$$V_B = 0 \text{ volt}$$

$$i_{2\Omega} = i_1 = \frac{V_A}{2} = \frac{2}{2} = 1 \text{ amp}$$



Nodal Analysis

Ex.6 Using Nodal Analysis Find Current in 8 ohm Resistance. AKTU-2021-22

Apply KCL at Node

$$i_1 + i_2 + 2 + i_3 = 0$$

$$\frac{90 - V_1}{20} + \frac{60 - V_1}{12} + \left(\frac{-V_1}{8} \right) = -2$$

$$V_1 \left[\frac{-1}{20} - \frac{1}{12} - \frac{1}{8} \right] + \frac{90}{20} + \frac{60}{12} = -2$$

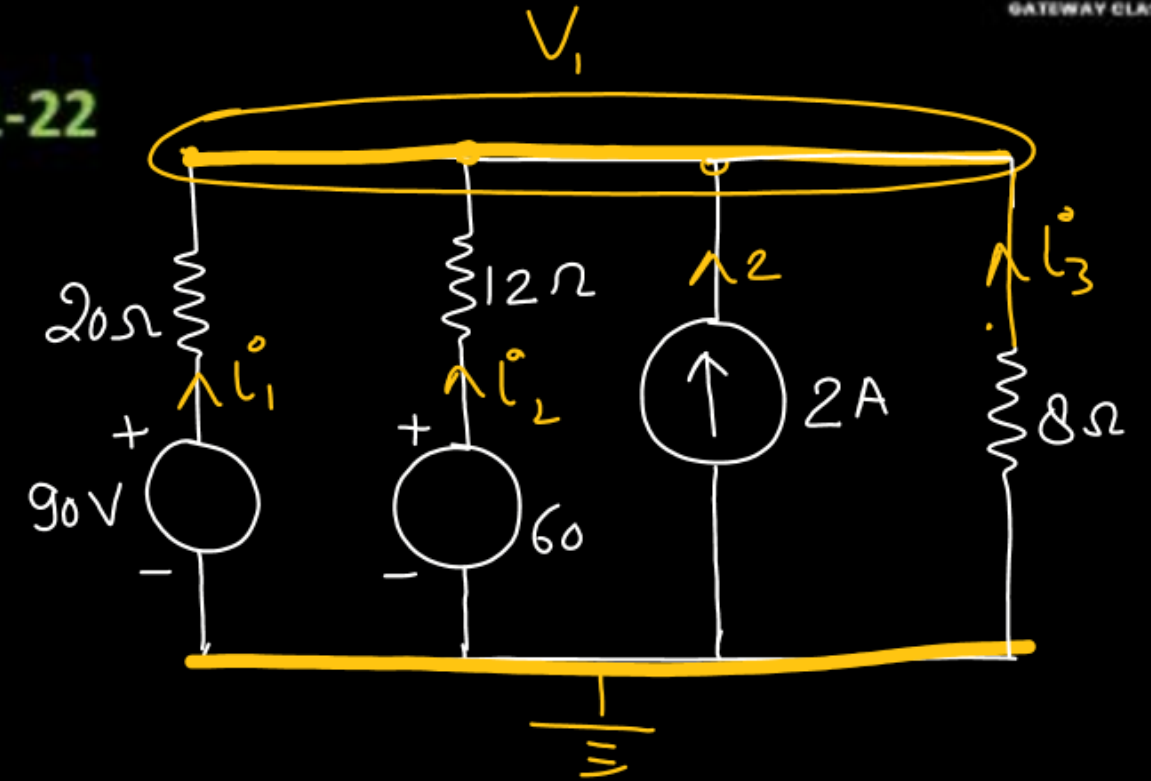
$$V_1 \left[\frac{-96 - 160 - 240}{1920} \right] = -2 - \frac{9}{2} - \frac{60}{12}$$

$$\neq V_1 [0.258] = -11.5$$

$$\boxed{V_1 = +44.57}$$

$$i_{8\Omega} = i_3 = \frac{-(44.57)}{8} = -5.5 \text{ Amp}$$

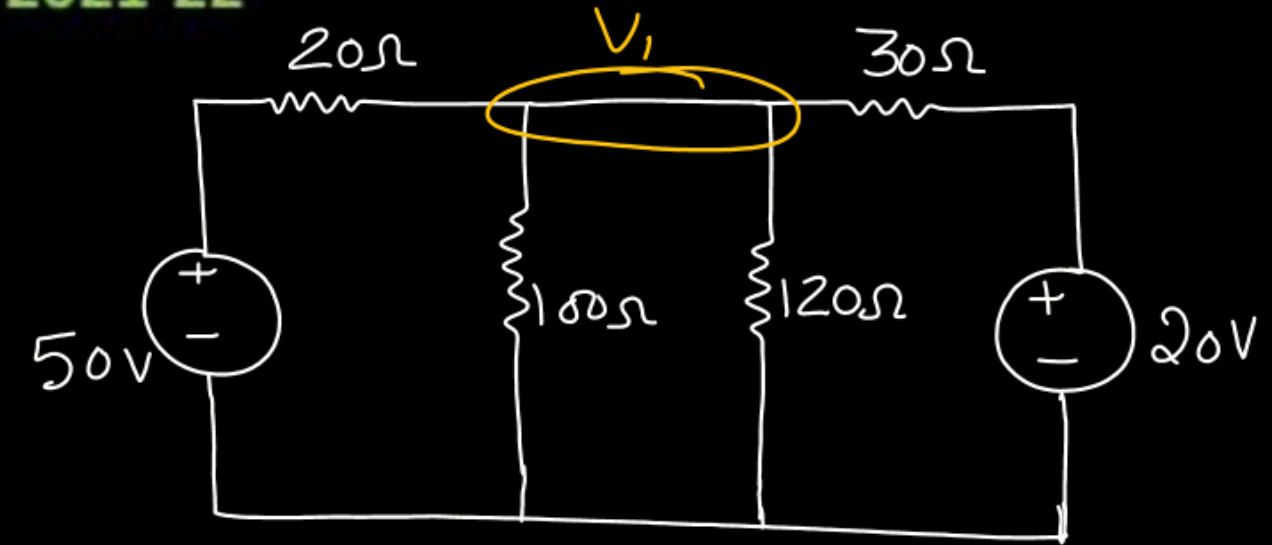
Ans:-



Ex.7 Using Nodal Analysis Find Current in r_2 Resistance.

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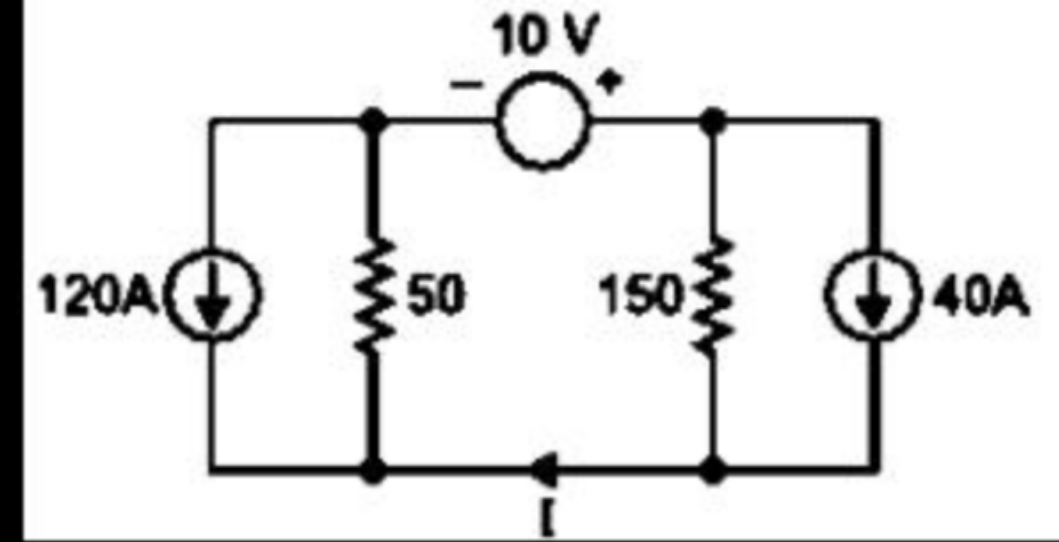
Practice Question for Home work.



DPP on Nodal Analysis

Q.1 Using Nodal Analysis Find Current in 50 ohm. **AKTU- 2021-22**

Ans = 120.05 Amp



Q.2 Using Nodal Analysis Find Current in 50 ohm. **AKTU- 2021-22, AKTU 2022-23**

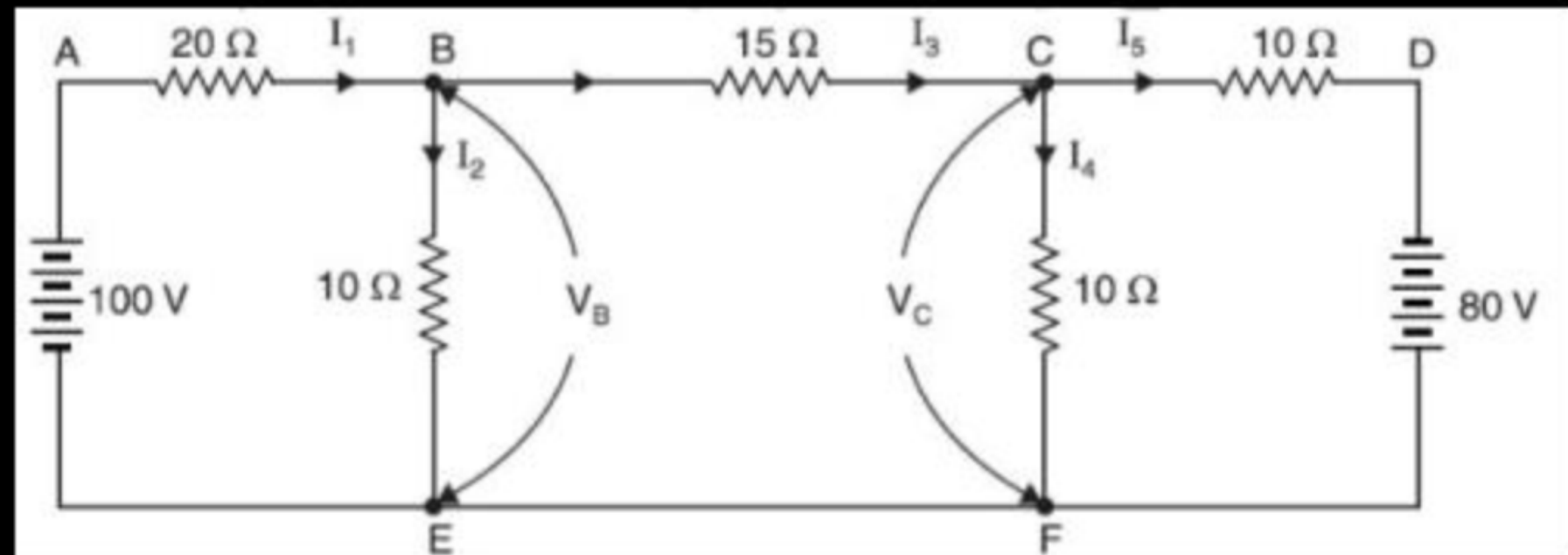
Ans. $I_1 = 4.25$ Amp.

$I_2 = 1.5$ amp ,

$I_3 = 2.75$ Amp

$I_4 = - 2.625$ amp

$I_5 = 5.37$ Amp.



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